

12-Month Software Engineer Roadmap

Goal: Become a strong full-stack software engineer and prepare for high-paying product company roles.

Phase 1 (Month 1-2): Core Programming

JavaScript Deep Dive: Execution Context, Scope, Closures, Hoisting, Event Loop, Promises, Async/Await, ES6+.

DSA: Arrays, Strings, Hash Maps, Linked Lists, Stacks, Queues. Target: 100 problems.

Phase 2 (Month 3-4): Advanced DSA

Trees, BST, Heaps, Graphs, Recursion, Backtracking, Dynamic Programming. Target: 300 total solved problems.

Phase 3 (Month 5): Frontend Mastery

React Hooks, Context API, Redux Toolkit, React Query, Performance Optimization, Next.js.

Phase 4 (Month 6-7): Backend Mastery

Node.js, Express, REST APIs, Authentication, Authorization, MongoDB, PostgreSQL.

Phase 5 (Month 8): System Design

SOLID Principles, Design Patterns, Load Balancing, Caching, Database Scaling, Message Queues.

Phase 6 (Month 9): DevOps & Cloud

Git, Docker, AWS (EC2, S3, RDS), GitHub Actions, CI/CD.

Phase 7 (Month 10): Resume Projects

Build School ERP, Library Management, Bus Ticket Booking, and E-commerce Platform.

Phase 8 (Month 11): Interview Preparation

DSA Revision, JavaScript, React, Node.js, DBMS, OS, Networking, Mock Interviews.

Phase 9 (Month 12): Job Applications

Apply to product companies and startups. Prepare resume and portfolio.

Daily Schedule (6 Hours)

2 Hours: DSA

2 Hours: Development

1 Hour: System Design

1 Hour: Interview Preparation